# Kota Fujimoto

Project Researcher

*Kyoto Institute of Technology, Kyoto, Japan*

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## Education

**The University of Tokyo** *Tokyo, Japan*

Ph.D. in Electrical Engineering and Information Systems *2026*

- Advisor: Dr. Hiroshi Fujimoto

**The University of Tokyo** *Tokyo, Japan*

M.S. in Advanced Energy *2023*

- Advisor: Dr. Hiroshi Fujimoto

**The University of Tokyo** *Tokyo, Japan*

B.E. in Aeronautics and Astronautics *2021*

- Advisor: Dr. Koji Uenishi

## Appointments

**KYOTO Design Lab, Center for the Possible Futures, Kyoto Institute of Technology** *Kyoto, Japan*

Project Researcher *Apr. 2026 – present*

**Ishizawa Lab, Institute of Industrial Science, The University of Tokyo** *Tokyo, Japan*

Visiting Researcher *Apr. 2026 – present*

**e-Mobility and Control Lab, Graduate School of Frontier Sciences, The University of Tokyo** *Tokyo, Japan*

Visiting Researcher *Apr. 2026 – present*

## Publications

### Published

Yifan Xu, Clara Cheung, Ming Shan Ng, Akilu Yunusa Kaltungo, Tsukasa Ishizawa, **Kota Fujimoto**, "Construction robotics: a systematic review of robot types, applications and human robot collaborations," *Results in Engineering*, Vol. 30, 110406, 2026.

**Kota Fujimoto**, Hiroshi Fujimoto, Yoshihiro Isaoka, Yuki Terada, "High Precision Control for Twin-Drive Machine Tool Based on Mode Decoupling with Virtual Viscosity," *IEEE/ASME Transactions on Mechatronics*, Vol. 30, No. 4, pp. 3054-3062, 2025.

Takuma Katagiri, Kentaro Yokota, **Kota Fujimoto**, Sakahisa Nagai, Hiroshi Fujimoto, "Wind Tunnel Verification of Velocity Control in Wing Coordinate System Using Acceleration-Based Disturbance Observer for Tilt-Wing eVTOL," *IEEJ Journal of Industry Applications*, Vol. 14, No. 2, pp. 188-195, 2025.

## Conference and Presentation

### Peer-reviewed

Yifan Xu, Clara Cheung, Ming Shan Ng, Akilu Yunusa Kaltungo, Tsukasa Ishizawa, **Kota Fujimoto**, "Analysing Construction Robotics Safety Standards for Safety Modelling and Simulation," *in International Symposium on Automation and Robotics in Construction (ISARC)*, pp. 810-817, 2026.

Ami Kumaki, **Kota Fujimoto**, Hiroshi Fujimoto, Yoshihiro Isaoka, Masataka Sakamoto, Yuki Terada, "Jerk-Timing-Based Polynomial Trajectory Optimization for Vibration Suppression," *in IEEE 19th International Conference on Advanced Motion Control (AMC)*, Daegu, South Korea, 2026.

Yusei Fujishima, **Kota Fujimoto**, Hiroshi Fujimoto, Yoshihiro Isaoka, Masataka Sakamoto, Yuki Terada, "Modeling and Analysis of Coupled System for Circular Motion in Machine Tools with XY-Axis Stage," *in IEEE 19th International Conference on Advanced Motion Control (AMC)*, Daegu, South Korea, 2026.

Yuki Tanaka, **Kota Fujimoto**, Sakahisa Nagai, Hiroshi Fujimoto, "Altitude Optimization above Transmitter Coil for In-flight Inductive Power Transfer Using Dynamic Programming," *in IEEE 19th International Conference on Advanced Motion Control (AMC)*, Daegu, South Korea, 2026.

Mayuko Hirota, Manto Kamiya, Yuto Naoki, **Kota Fujimoto**, Kentaro Yokota, Binh-Minh Nguyen, Sakahisa Nagai, Osamu Shimizu, Hiroshi Fujimoto, "Inertia Estimation About z-Axis Using Motor Torque for Multi-Rotors," *in The 12nd IEEJ international workshop on Sensing, Actuation, Motion Control, and Optimization (SAMCON)*, Daegu, Korea, 2026.

**Kota Fujimoto**, Hiroshi Fujimoto, Yoshihiro Isaoka, Yuki Terada, "Practical Implementation of High Precision Control Based on Mode Decoupling with Virtual Viscosity for Parallel Twin-Drive Machine Tool System," *in The 12nd IEEJ international workshop on Sensing, Actuation, Motion Control, and Optimization (SAMCON)*, Daegu, Korea, 2026.

Shikauchi Yutaka, **Kota Fujimoto**, Osamu Shimizu, Hiroshi Fujimoto, Shuji Kawano, "Verification of Big Data Analysis on Wireless Power Transfer For Electric Vehicles Focused on Traffic Signal GPS," *in 7th International Electric Vehicle Technology Conference 2025 (EVTeC)*, Yokohama, Japan, 2025.

**Kota Fujimoto**, Hiroshi Fujimoto, Yoshihiro Isaoka, Yuki Terada, "High Precision Control for Parallel Twin-Drive Machine Tool System Based on Offline Parameter Identification Focusing on Differential Mode," *in The 11st IEEJ international workshop on Sensing, Actuation, Motion Control, and Optimization (SAMCON)*, Shimane, Japan, 2025.

Takuma Katagiri, **Kota Fujimoto**, Sakahisa Nagai, Hiroshi Fujimoto, "Trajectory Tracking Control of Tilt-Wing eVTOL Using Aerodynamic Coupling Observer and Wing Angle Correction Method," *in The 2025 IEEE International Conference on Mechatronics (ICM)*, Wollongong, Australia, 2025.

Pinzhe Feng, Yuto Naoki, **Kota Fujimoto**, Kentaro Yokota, Sakahisa Nagai, Hiroshi Fujimoto, "Basic Study on Short Distance Landing Control with Reverse Thrust of Variable Pitch Propeller for Fixed Wing Electric Aircraft," *in The 2025 IEEE International Conference on Mechatronics (ICM)*, Wollongong, Australia, 2025.

**Kota Fujimoto**, Osamu Shimizu, Hiroshi Fujimoto, Jin Katsuya, Tetsu Sato, "Feasibility Study for Dynamic Wireless Power Transfer in Simulation with Big Data Focusing on Battery Capacity Requirement of Electric Vehicles," *in the 27th edition of the International Symposium on Power Electronics, Electrical Drives, Automation and Motion (SPEEDAM)*, Ischia, Italy, 2024.

Sakahisa Nagai, Yusuke Satoh, **Kota Fujimoto**, Hiroshi Fujimoto, "Sensorless Drone Detection for In-Flight Inductive Power Transfer Using Transmitter-Side Voltage Pulses," *in IEEE 2024 Wireless Power Technology Conference and Expo (WPTCE)*, Kyoto, Japan, 2024.

Takuma Katagiri, Kentaro Yokota, **Kota Fujimoto**, Sakahisa Nagai, Hiroshi Fujimoto, "Basic Study on Velocity Control in Wing Coordinate System Using Acceleration-based Disturbance Observer for Tilt-wing eVTOL," *in The 10th IEEJ international workshop on Sensing, Actuation, Motion Control, and Optimization (SAMCON)*, Kyoto, Japan, 2024.

Shogo Yamada, **Kota Fujimoto**, Hiroshi Fujimoto, "Basic Study on Robust Perfect Tracking Control Based on Parameter Estimation Reducing Outlier Effects," *in The 10th IEEJ international workshop on Sensing, Actuation, Motion Control, and Optimization (SAMCON)*, Kyoto, Japan, 2024.

**Kota Fujimoto**, Hiroshi Fujimoto, Alessandro Correa Victorino, Pedro Castillo, "Optimal Energy Trajectory Generation Based on Pitch-Dependent Mutual Inductance Model for In-Flight Inductive Power Transfer of Drones," *in IEEE 18th International Conference on Advanced Motion Control (AMC)*, Kyoto, Japan, 2024.

**Kota Fujimoto**, Hiroshi Fujimoto, Yoshihiro Isaoka, Yuki Terada, "High Precision Control for Twin-Drive System Based on Mode Decoupling with Virtual Viscosity: Equivalent Controller Transform for Machine Tool," *in IEEE 18th International Conference on Advanced Motion Control (AMC)*, Kyoto, Japan, 2024.

**Kota Fujimoto**, Sakahisa Nagai, Binh-Minh Nguyen, Hiroshi Fujimoto, "Test Bench Study on Attitude Estimation in Ground Effect Region Based on Motor Current for In-Flight Inductive Power Transfer of Drones," *in IEEE 2023 The 49th Annual Conference of the IEEE Industrial Electronics Society (IECON)*, Singapore, 2023.

Yusuke Satoh, **Kota Fujimoto**, Ryo Matsumoto, Sakahisa Nagai, Binh-Minh Nguyen, Hiroshi Fujimoto, "Basic Study on Received Power Control of In-Flight Inductive Power Transfer for Drones by Active Rectifier Switching and Altitude Regulation," *in IEEE 2023 The 49th Annual Conference of the IEEE Industrial Electronics Society (IECON)*, Singapore, 2023.

**Kota Fujimoto**, Takumi Hamada and Hiroshi Fujimoto, "Proposal on Model Based Current Overshoot Suppression of Receiver Side Coil in Drone Wireless Power Transfer System," *in IEEE 2022 Wireless Power Week (WPW)*, pp. 235-239, Bordeaux, France, 2022.

### Non-peer-reviewed

Ami Kumaki, **Kota Fujimoto**, Hiroshi Fujimoto, Masataka Sakamoto, Yoshihiro Isaoka, Yuki Terada, "Trajectory Generation under Speed-Dependent Acceleration Constraints for Residual Vibration Suppression through Deceleration Jerk Design: Application to Machine Tools," *in IEE-Japan Technical Meeting on Industrial Instrumentation and Control (PSS)*, No. IIC-26-016, pp. 89-94, Tokyo, Japan, 2026.

Yusei Fujishima, **Kota Fujimoto**, Hiroshi Fujimoto, "Proposal of Physical-coordinate Additive Perfect Tracking Control Focusing on Rigid-body Modes in Torsional Vibration System," *in IEE-Japan Technical Meeting on Industrial Instrumentation and Control (PSS)*, No. IIC-26-011, pp. 61-66, Tokyo, Japan, 2026.

Ten Arai, Maho Matsukura, Daigo Takasaki, Isamu Moriai, Kenshin Yamakawa, Keisuke Yorita, **Kota Fujimoto**, Hiroyuki Koizumi, Kimiya Komurasaki, "Reduction of Electromagnetic Disturbances in a Wireless Power Transfer System Used in Thrust Stands for Micro Propulsion Systems," *in The 65th Conference on Aerospace Propulsion and Power*, Oita, Japan, 2026.

Mayuko Hirota, Manto Kamiya, Yuto Naoki, **Kota Fujimoto**, Binh-Minh Nguyen, Sakahisa Nagai, Osamu Shimizu, Hiroshi Fujimoto, Kentaro Yokota, "Basic Study on Z-axis Inertia Estimation Method Using Counter-torque Observer in Multirotors," *in IEE-Japan Technical Meeting on Mechatronics Control (PSS)*, No. MEC-25-007, Tokyo, Japan, 2025.

**Kota Fujimoto**, Hiroshi Fujimoto, Yoshihiro Isaoka, Yuki Terada, "High Precision Control for Twin-Drive System of Machine Tool Based on Mode Decoupling with Virtual Viscosity: Feedforward Realization of Virtual Viscosity on Two-Inertia System," *in IEE-Japan Technical Meeting on Mechatronics Control (PSS)*, No. MEC-24-012, Tokyo, Japan, 2024.

**Kota Fujimoto**, Sakahisa Nagai, Binh-Minh Nguyen, Hiroshi Fujimoto, "Motor Current Based Attitude Estimation Considering Ground Effect for In-flight Inductive Power Transfer of Drones: Basic Concept and Test-Bench Evaluation," *in IEE-Japan 2024 Industry Applications Society Conference (JIASC)*, No. 1-60, Ibaraki, Japan, 2024.

**Kota Fujimoto**, Yusuke Satoh, Sakahisa Nagai, Hiroshi Fujimoto, "Basic Study on Receiving-Side System Design of In-flight Inductive Power Transfer for Drones Considering Weight Limitation," *in IEE-Japan Joint Technical Meeting on Power Engineering/Power Systems Engineering/Semiconductor Power Converter (SPC)*, No. PE-24-059,PSE-24-071,SPC-24-113, Nagasaki, Japan, 2024.

**Kota Fujimoto**, Hiroshi Fujimoto, Yoshihiro Isaoka, Yuki Terada, "High Precision Control for Twin-Drive System of Machine Tool Based on Mode Decoupling with Virtual Viscosity: Basic Study on Two-Inertia System," *in IEE-Japan Technical Meeting on Mechatronics Control (PSS)*, No. MEC-23-011, Tokyo, Japan, 2023.

Yusuke Satoh, **Kota Fujimoto**, Ryo Matsumoto, Sakahisa Nagai, Binh-Minh Nguyen, Hiroshi Fujimoto, "Basic Study on Flight Altitude and Battery Current Control in In-Flight Wireless Power Transfer for Drones," *in IEE-Japan Joint Technical Meeting on Power Engineering/Power Systems Engineering/Semiconductor Power Converter (SPC)*, No. PE-23-084,PSE-23-090,SPC-23-140, Okinawa, Japan, 2023.

**Kota Fujimoto**, Osamu Shimizu, Hiroshi Fujimoto, Jin Katsuya, Tetsu Sato, "Requirement Analysis for Dynamic Wireless Power Transfer with Big Data from Real Vehicles," *in IEE-Japan Joint Technical Meeting on Power Engineering/Power Systems Engineering/Semiconductor Power Converter (SPC)*, No. SPC23140, Okinawa, Japan, 2023.

**Kota Fujimoto**, Kentaro Yokota, Sakahisa Nagai, Hiroshi Fujimoto, "Modeling and Verification of Pitch-Dependent Coupling Coefficient for WPT to Flying Drone," *in IEE-Japan 2021 Industry Applications Society Conference (JIASC)*, No. 48, Niigata, Japan, 2021.

## Awards, Fellowships, & Grants

### Awards

2025 **2025 IEEJ Industry Applications Society Excellent Presentation Award, The Institute of Electrical Engineers of Japan**, The Institute of Electrical Engineers of Japan

2022 **APET Excellent Poster Presentation Award 3rd Prize**, Alliance for Advanced Power & Environmental Technology

2021 **IEEJ Excellent Presentation Award**, The Institute of Electrical Engineers of Japan

### Fellowship

2023-2025 **Fostering Advanced Human Resources to Lead Green Transformation (SPRING GX)**, The University of Tokyo

### Grant

2025 **The Dispatch for International Meetings Program (for IEEE AIM2025)**, Fostering Advanced Human Resources to Lead Green Transformation (SPRING GX)
2024 **The Dispatch for International Meetings Program (for IEEE IECON2023)**, Fostering Advanced Human Resources to Lead Green Transformation (SPRING GX)
2023 **Overseas Travel Expense Assistance**, The Telecommunications Advancement Foundation
2022-2023 **La Bourse de Mobilité Internationale de Sorbonne Universités**, Sorbonne Universités
2022 **Researcher Overseas Training Grant**, Suzuki Foundation

## Teaching/Supervising Experience

2021-2025 **Graduate Teaching Assistant**, Electrical and Electronic Engineering, The University of Tokyo

## Outreach & Professional Development

### Service and Outreach

2022-2025 **School Festival**, Operational Stuff at Lab Introduction exhibit *Tokyo*
2022 **Workshop in Kashiwanoha innovation festival**, Operational Stuff at Craft Construction Course of Dynamic Wireless Power Transfer *Chiba*

### Internship

2024 May-Aug. **Université de technologie de Compiègne**, Project internship student *France*
2023 Aug.-Nov. **Université de technologie de Compiègne**, Project internship student *France*
2022 **ACSL Ltd.**, R&D Mechatronics team *Tokyo*

### Professional Memberships

The Institute of Electrical and Electronics Engineers Member
The Institute of Electrical Engineers of Japan Member
The Japan Society for Aeronautical and Space Sciences Member
Society of Automotive Engineers of Japan Member